

Vector Spaces

Vector spaces are denoted by $\mathbb{R}^1, \mathbb{R}^2, \mathbb{R}^3, \dots, \mathbb{R}^n$; the space $\mathbb{R}^n$ consists of all column vectors with n components.

- i. Addition of any two vectors is in vector space
- ii. Multiplication by scalar is in vector space

Examples

$\mathbb{R}^1$ = all 1-dimensional real vectors
= a line

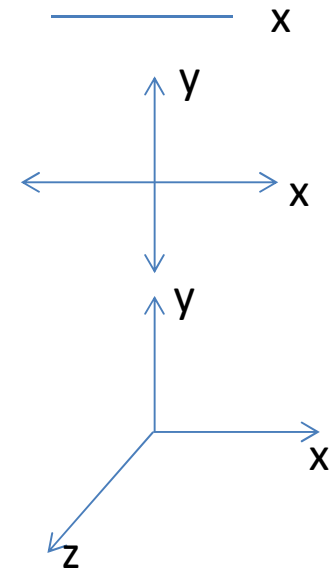
$[-2], [0], [3], \dots$

$\mathbb{R}^2$ = all 2-dimensional real vectors
= x-y plane

$\begin{bmatrix} -3 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1.12 \\ 4.01 \end{bmatrix}, \dots$

$\mathbb{R}^3$ = all 3-dimensional real vectors
= x-y-z plane

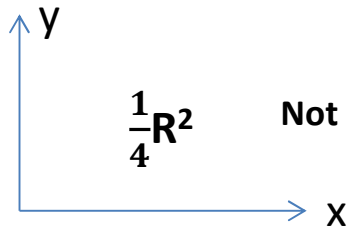
$\begin{bmatrix} -3 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1.12 \\ 4.01 \\ 0.53 \end{bmatrix}, \dots$



Not a vector space:

$\frac{1}{4}\mathbb{R}^2$

Not closed by multiplication with scalar



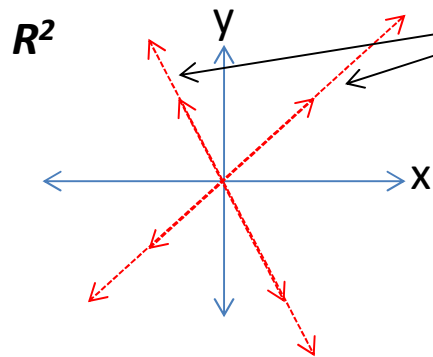
Vector Spaces

Subspace: a subspace of a vector space is a nonempty subset that satisfies the requirements for a vector space.

- i. If we add any vectors x and y in the subspace, $x + y$ is in the subspace.
- ii. If we multiply any vector x in the subspace by any scalar c , cx is in the subspace.

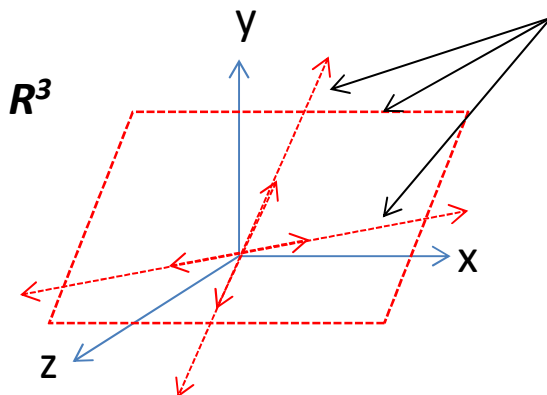
i.e., linear combinations stay in the subspace.

Examples:



Subspace of R^2 : A line in R^2 passing through $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is a subspace of vector space R^2 .

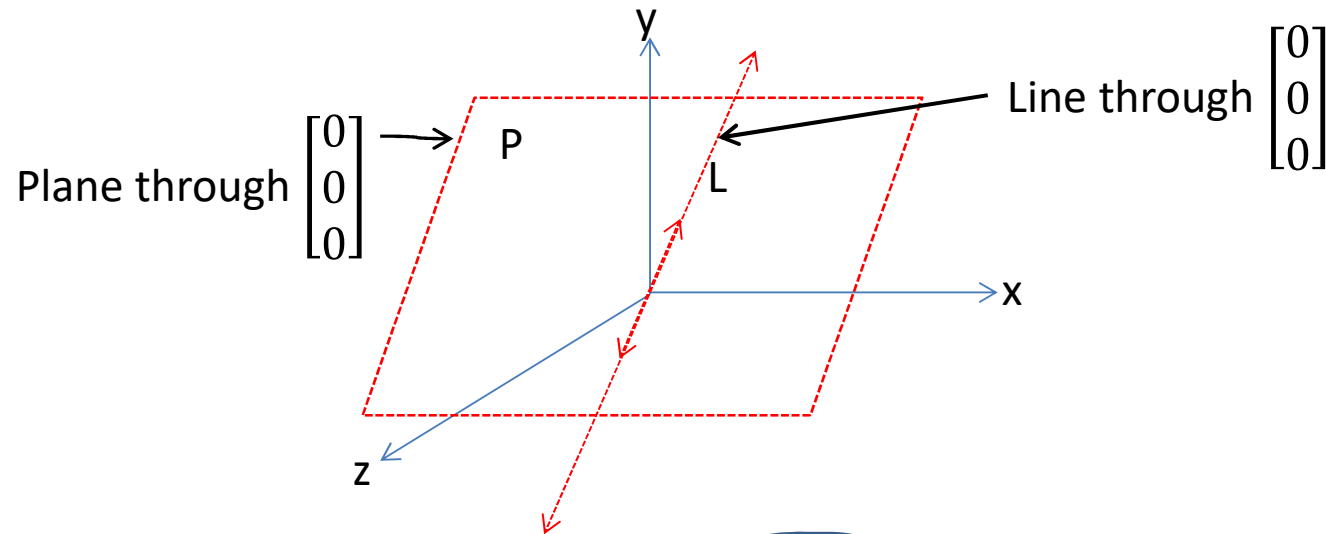
- i. R^2
- ii. Any line through $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.
- iii. $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$



Subspace of R^3 : A line in R^3 passing through $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a subspace of vector space R^3 .

- i. R^3
- ii. Any line through $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
- iii. $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
- iv. Any plane through $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Vector Spaces



Union of two subspaces P and L:

$P \cup L =$ all vectors in P or L or both

This is not a subspace of R^3

Intersection of two subspace P and L:

$P \cap L =$ all vectors in both P and L

It may be $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ or line L lie on the plane P

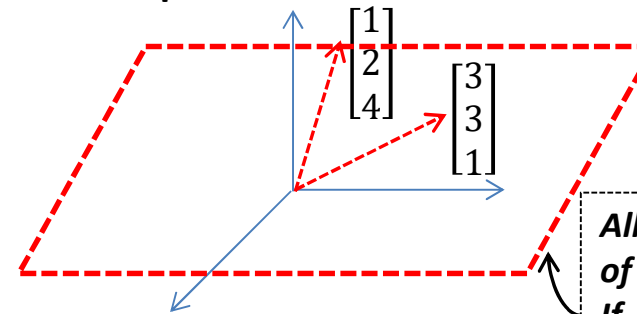
This is a subspace of R^3

Column space and null space of a matrix

Column Space: The column space of a $m \times n$ matrix A contains all linear combinations of the columns of A . It is a subspace of R^m .

Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$, The column space of A denoted as $C(A)$ is a subspace of R^3 , which is obtained from all the

linear combinations of column vectors $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}$.



All linear combinations of col 1 and col 2 of A . If two columns are same vector then $C(A)$ will be a LINE

To solve $Ax = b$:

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow u \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + v \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

The system $Ax = b$ is solvable if and only if the vector b can be expressed as a combination of the columns of A . i.e., b is in the column space.

Example: Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$, (i) What will be the column space of A ?

Column space and nullspace of a matrix

Answer: The column space $C(A)$ is subspace of R^4 . $C(A)$ is the set of vectors contains all the linear combinations of the columns vectors of A . Since, the 3rd column is a linear combination of 1st and 2nd columns, it is not linearly independent from 1st and 2nd columns, i.e., the linear combinations of 1st, 2nd and 3rd columns of A is same plane as linear combination of 1st and 2nd columns. Thus, the column space of A is a 2-dimensional subspace of R^4 .

(ii) Does $Ax = b$ have a solution for every b ?

$$\text{Answer: } Ax = b \Rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \Rightarrow u \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + v \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + w \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

Since, the column space of A is a 2-dimensional subspace of R^4 . It is not possible to have a solution of $Ax = b$ for every b .

(iii) Which b 's allows this system to be solved?

$$\text{Answer: } b = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 6 \\ 8 \\ 10 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ etc.}$$

I can solve $Ax = b$ exactly when b is in the column space $C(A)$.

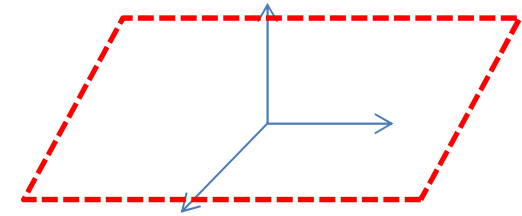
Column space and nullspace of a matrix

Nullspace: *The null space of a matrix consists of all vectors X such that $AX=0$. It is denoted by $N(A)$. It is a subspace of R^n , just as the column space was a subspace of R^m .*

Example: Find the nullspace of $A = \begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 4 \end{bmatrix}$

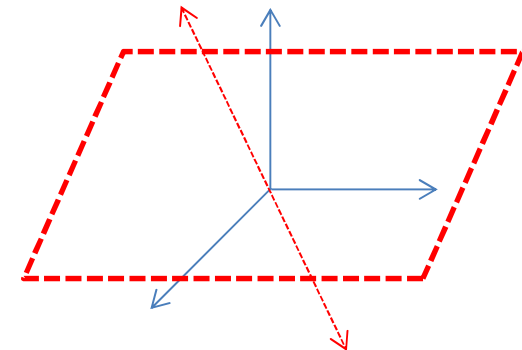
Answer: $AX = 0 \Rightarrow \begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$

For $u = 0$ and $v = 0$, $AX = 0$. So, the nullspace contains only the vector $(0,0)$. This matrix has “independent columns”



Example: Find the nullspace of $B = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix}$

Answer: $AX = 0 \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$



So the nullspace of B contains the vector $(1, 1, -1)$ and automatically contains any multiple $(c, c, -c)$. The nullspace of B is the line of all points $u = c, v = c, w = -c$. This matrix has one “dependent column”

The solutions of $AX = 0$ always give a subspace i.e., nullspace is a subspace:

Proof: Let V and W be two vector are in nullspace of A . So, $AV = 0$ and $AW = 0$. Now $A(V + W) = AV + AW = 0$, i.e., $V + W$ is in $N(A)$. If $AV = 0$, then $A(cV) = 0$, i.e., all linear combination of V and W are in $N(A)$. Thus, Nullspace is a subspace.

Note: when $AX = b$ and $b \neq 0$, X does not guaranteed it is a subspace.

Solving $AX=0$

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \text{ and } AX = 0, \text{ i. e., } \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

By applying forward elimination:

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow U$$

Pivot col
Free col
Pivot col
Free col

This column is a combination of earlier columns (pointing to the first column of the first matrix)

This row is a combination of earlier rows (pointing to the third row of the second matrix)

Rank of a matrix = No. of pivot. So rank of $A = 2$. I can assign any value to the free variables x_2 and x_4 .

Let $x_2 = 1, x_4 = 0$. Then the values of x_1, x_3 of the following linear system

$$\begin{aligned} x_1 + 2x_2 + 2x_3 + 2x_4 &= 0 \\ 2x_3 + 4x_4 &= 0 \end{aligned} \quad \text{will be } x_1 = -2, x_3 = 0.$$

So the bunch of solution (known as special solution) for $AX=0$ will be $X = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, where c is a scalar.

Let $x_2 = 0, x_4 = 1$. Then by back-substitution we will get $x_1 = 2, x_3 = -2$. So another bunch of special

solution for $AX=0$ will be $X = d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$, where d is a scalar.

Solving $AX=0$

So, $X = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$ is the nullspace of $AX = 0$, i.e., $UX = 0$. Thus, nullspace contains all the combinations of special solutions. If rank of a matrix is r (no. of pivot), then no. of free variables is $(n - r)$. So nullspace will be the combination of the $(n - r)$ special solutions.

Reduced row Echelon Form

$$U = \begin{bmatrix} \bullet & * & * & * & * & * & * & * \\ 0 & \bullet & * & * & * & * & * & * \\ 0 & 0 & 0 & \bullet & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \bullet \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 0 & * & 0 & * & * & * & 0 \\ 0 & 1 & * & 0 & * & * & * & 0 \\ 0 & 0 & 0 & 1 & * & * & * & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The entries of a 5 by 8 echelon matrix U and its reduced form R .

Method: $U = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{zero above and below the pivot}} \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{zero above and below the pivot}} \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\begin{matrix} & & & R \\ \text{Pivot} & \text{Free} & \text{Pivot} & \text{Free} \end{matrix}$

Let $x_2 = 1, x_4 = 0$. From the linear system $x_1 + 2x_2 - 2x_4 = 0, x_3 + 2x_4 = 0$ we will get

$x_1 = -2, x_3 = 0$. So, for free variables $x_2 = 1, x_4 = 0, X = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$.

Solving $AX=0$, Reduced Row Echelon Form

Let $x_2 = 0, x_4 = 1$. From the linear system $x_1 + 2x_2 - 2x_4 = 0, x_3 + 2x_4 = 0$ we will get $x_1 = 2, x_3 = -2$. So, for free variables $x_2 = 0, x_4 = 1$, $X = d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$. Thus, $X = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$ is the nullspace of

$AX = 0$ or $UX = 0$ or $RX = 0$.

Note one can directly get the values of pivot variables x_1 , and x_3 from the reduced row matrix R without solving linear system as done above. After derive the reduced row matrix R, all the special solutions are directly obtained from the free columns of R.

Example: Find the nullspace of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix}$

Solving $AX=b$, Reduced Row Echelon Form

To find complete solution to $AX=b$

Step 1: Compute X -Particular (X_P)

- Set all free variables to zero.
- Solve $AX=b$, for pivot variables

Step 2: Compute nullspace (X_N)

Step 3: Compute the Complete solution $X = X_P + X_N$. Since $AX_P = b$ and $AX_N = 0$, $A(X_P + X_N) = b + 0 = b$. So, $AX = b$.

Example: Solve the following linear system:

$$x_1 + 2x_2 + 2x_3 + 2x_4 = 1$$

$$2x_1 + 4x_2 + 6x_3 + 8x_4 = 5$$

$$3x_1 + 6x_2 + 8x_3 + 10x_4 = 6$$

Answer: Applying forward elimination:

$$\begin{bmatrix} 1 & 2 & 2 & 2 & 1 \\ 2 & 4 & 6 & 8 & 5 \\ 3 & 6 & 8 & 10 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 & 1 \\ 0 & 0 & 2 & 4 & 3 \\ 0 & 0 & 2 & 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 & 1 \\ 0 & 0 & 2 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivot Free Pivot Free

Now assuming free variables $x_2 = 0, x_4 = 0$, the above linear system simplified as:

$$x_1 + 2x_3 = 1$$

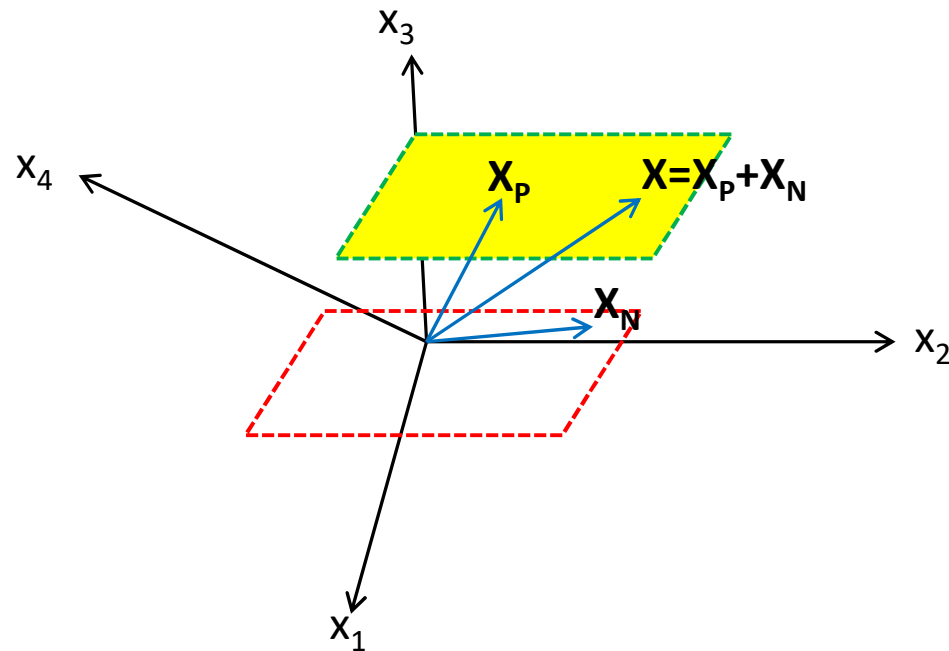
$$2x_3 = 3$$

and $x_3 = \frac{3}{2}, x_1 = -2$. So, $X_P = \begin{bmatrix} -2 \\ 0 \\ 1.5 \\ 0 \end{bmatrix}$.

Solving $AX=b$, Reduced row Echelon Form

Now we need to compute the nullspace of matrix A. Which is $X_N = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$. So the

$$\text{complete solution } X = X_P + X_N = \begin{bmatrix} -2 \\ 0 \\ 1.5 \\ 0 \end{bmatrix} + c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}.$$



Plot of all solutions X in $\mathbb{R}^4$

Rank of Matrix

Elimination reveals the pivot variables and free variables. If there are r pivots, there are r pivot variables and $n-r$ free variables. That important number r will be given a name—it is the rank of the matrix.

Suppose elimination reduces $Ax = b$ to $Ux = c$ and $Rx = d$, with r pivot rows and r pivot columns. The rank of those matrices is r . The last $m - r$ rows of U and R are zero, so there is a solution only if the last $m-r$ entries of c and d are also zero.

Full column rank matrix A (means, $r=n$ and no free variables):

- Nullspace of A is only Zero vector.
- Solution to $AX = b$ is $X = X_p$ (unique solution if it exist)

Example: $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 6 & 1 \\ 5 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 0 & -5 \\ 0 & -17 \\ 0 & -14 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 0 & -5 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = U \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = R$

So X have either 0 or 1 solution

Full row rank matrix A (means, $r=m$ and every row have a pivot):

- $n-r$ free variables, i.e., nullspace is not Zero.
- $Ax=b$ will be solved for every b

Example: $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 7 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 0 \end{bmatrix} = R$

So, $X = X_p + X_N = X_p + c$

$$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Pivot free

Infinite number of solutions

Rank of Matrix

Full rank square matrix A (means, $m=n=r$ and no free variables):

- Nullspace of A is only Zero vector.
- Invertible matrix.
- Solution to $AX = b$ is $X = X_p = A^{-1}b$

Exactly one solution

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = R$

So, $X = X_p$

Rank $< m$ and rank $< n$:

- $n-r$ free variables i.e., nullspace is not Zero.
- $Ax=b$ will not have solution for every b .

0 or ∞ solutions

Example: $A = \begin{bmatrix} 2 & 4 & 8 \\ 4 & 4 & 8 \\ 0 & 4 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 4 & 8 \\ 0 & -4 & -8 \\ 0 & 4 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 4 & 8 \\ 0 & -4 & -8 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} = R$

Since X_p have either no solution or infinitely many solutions, X have 0 or ∞ solutions.

Summery

I. $r = m = n$

$$R = [I]$$

One solution

II. $r = n < m$

$$R = \begin{bmatrix} I \\ 0 \end{bmatrix}$$

0 or One solution

III. $r = m < n$

$$R = [IF]$$

∞ solutions

IV. $r < m, r < n$

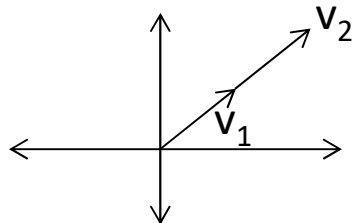
$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

0 or ∞ solutions

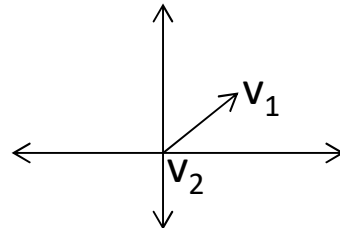
Linear Independence, Basis, and Dimension

Given a set of vectors $v_1, v_2, \dots, v_k$, suppose $c_1v_1 + \dots + c_kv_k = 0$ only happens when $c_1 = c_2 = \dots = c_k = 0$. Then the vectors $v_1, v_2, \dots, v_k$ are linearly independent. If any c 's are nonzero, the v 's are linearly dependent.

Examples:

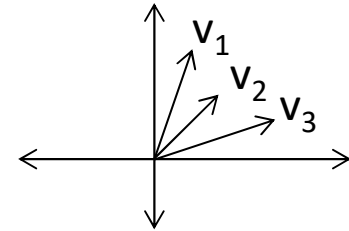


$v_2 = 2v_1$
 $v_2 - 2v_1 = 0$
 So, v_1, v_2 are dependent



$0 \cdot v_1 + c \cdot v_2 = 0$
 So, v_1, v_2 are dependent

$$A = \begin{matrix} & v_1 & v_2 & v_3 \\ \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \end{matrix}$$



Since there is a free column, nullspace is a line in R^3 . So the linear combination of v_1, v_2, v_3 is Zero vector. Thus, they are dependent.

Spanning a Space: Vectors $v_1, v_2, \dots, v_d$ span a space means, the vector space consists of all the linear combinations of those vectors.

Examples: The vectors $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ span the vector space R^2 .

The vector $v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $v_3 = \begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}$ span a plane (the x-y plane) in R^3 .

Basis for a vector space: A basis for a vector space is a sequence of vectors $v_1, v_2, \dots, v_d$ having two properties at once:

1. They are independent.
2. They span the vector space

Linear Independence, Basis, and Dimension

Examples: One basis of vector space R^3 is $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. $U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ *Rank=3, no free variable. Nullspace is only Zero vector. Thus, column vectors are basis of R^3*

Another basis of vector space R^3 is $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. $U = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$ *Rank=3, no free variable. Nullspace is only Zero vector. Thus, column vectors are basis of R^3*

Is $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 8 \end{bmatrix}$ is a basis of R^3 ? $U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ *Rank=3, no free variable=1. Nullspace is line in R^3 passing through origin. Thus, column vectors are not independent.*

Column vectors of any invertible $n \times n$ matrices are the basis of R^n

*Although there are many basis but they all have the **same number** of vectors. If we talk of R^3 , then number of vectors is 3. If we talk of R^n , then the number of vectors in the basis is n . Given a space every basis for that space has same number of vectors.*

Dimension of the space: Number of vectors in the basis is called the dimension of the space.

Example: What is the dimension of the column space of matrix $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$?

Linear Independence, Basis, and Dimension

Answer: $u = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Pivot Pivot Free Free

Column space is a 2-dimensional plane in $\mathbb{R}^3$. So one set of Basis of $C(A)$ is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$.

Similarly, other set of basis will be $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$ or $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 5 \end{bmatrix}$ etc. So many basis can be found in $C(A)$, But all have dimension 2. Dimension of $C(A)$ = Rank of A .

Example: What is the dimension of the nullspace of matrix $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$?

Answer: $R = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $X_N = c \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

So nullspace is a 2-dimensional place in $\mathbb{R}^4$. Dimension of nullspace $N(A)$ = No. of free columns = $n - r = 2$

The Four Fundamental Subspaces

Four fundamental subspaces of a matrix A are:

1. Column space $C(A)$
2. Null space $N(A)$
3. Row space (all combination of rows)
 $\Rightarrow$ All combination of columns of A^T
 $\Rightarrow C(A^T)$
4. Null space of A^T , $N(A^T)$

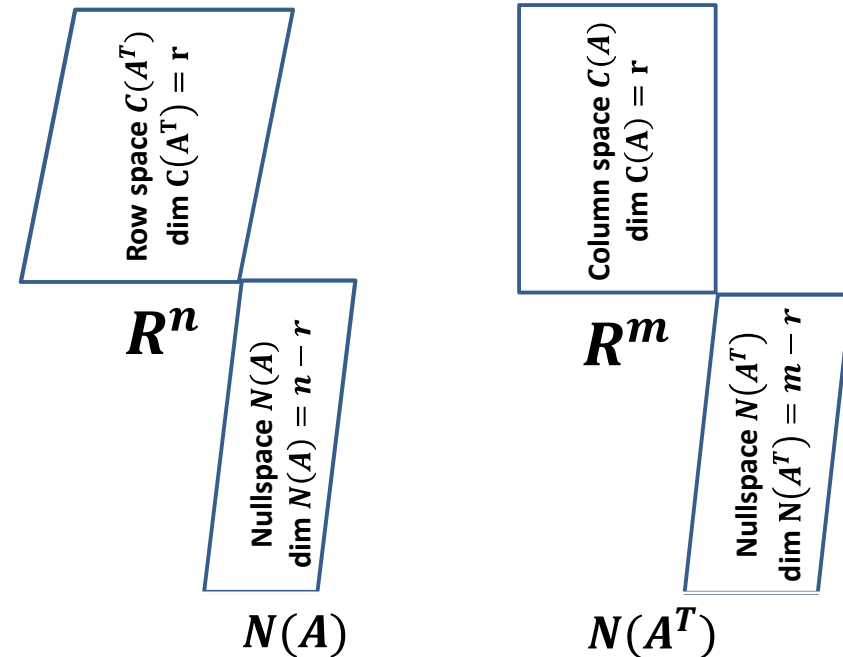
For $m \times n$, A :

$C(A)$ is subspace of $\mathbb{R}^m$

$N(A)$ is subspace of $\mathbb{R}^n$

$C(A^T)$ is subspace of $\mathbb{R}^n$

$N(A^T)$ is subspace of $\mathbb{R}^m$



	$C(A)$	$C(A^T)$	$N(A)$	$N(A^T)$
Basis	Pivot columns	Pivot rows	Special solution for free variables $n - r$	Special solution for free variables $m - r$
Dimension	r	r	$n - r$	$m - r$

Other space

Example: Find the four fundamental subspaces of $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}$.

Matrix space

Formed by linear combinations ($c \cdot A + d \cdot B, c \cdot A$) of matrices. Let M be the space of all 3x3 matrices. The subspace of M are:

- U =All 3x3 upper triangular matrices.
- S =All 3x3 symmetric matrices.
- D =All 3x3 diagonal matrices.

Basis for M : $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\text{Dim}(M) = 9$$

Basis for U : $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\text{Dim}(U) = 6$$

Basis for S : $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$$\text{Dim}(S) = 6$$

Basis for D : $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\text{Dim}(D) = 3$$